AWS Prescriptive Guidance
Agentic AI patterns and workflows on AWS

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Speech and voice agents

Speech and voice agents interact with users through spoken dialogue. These agents integrate speech recognition, natural-language understanding, and speech synthesis to enable conversational AI across telephony, mobile, web, and embedded platforms.

Voice agents are particularly effective in hands-free, real-time, or accessibility-driven environments. By combining streaming interfaces with LLM-powered reasoning, they facilitate rich, dynamic interactions that feel natural to users.

Architecture

A speech and voice agent is shown in the following diagram:

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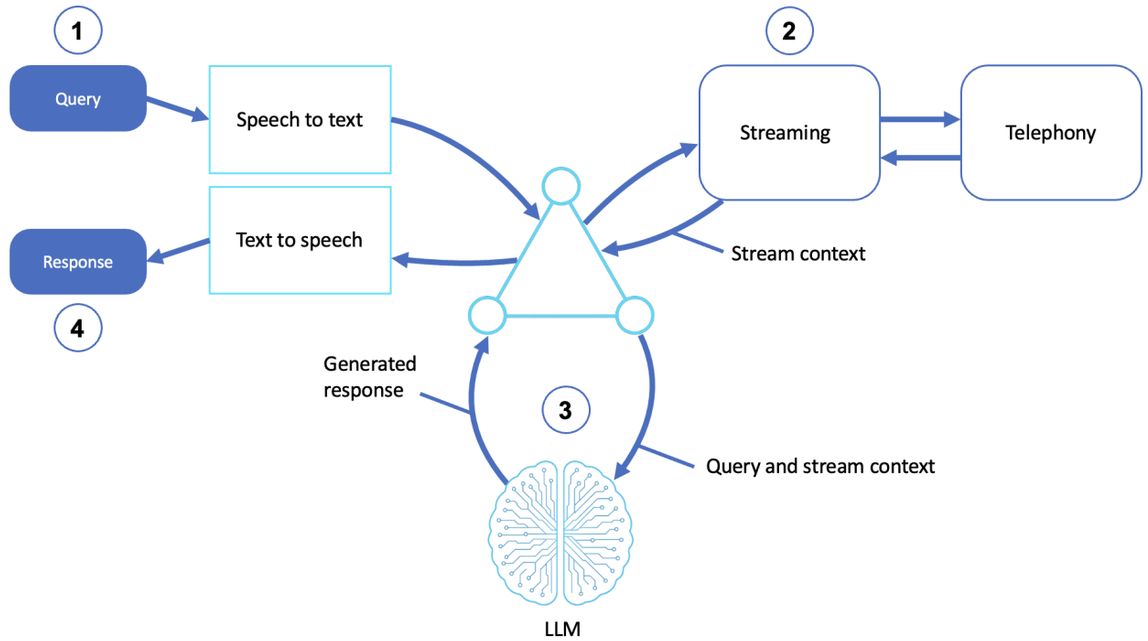

Description

1. Receives a voice query
   - The user voices a request to a phone, microphone, or embedded system.
   - A speech-to-text (STT) module converts the audio to text.
2. Integrates streaming and telephony context
   - The agent uses a streaming interface to manage audio I/O in real time.
   - If it's deployed in a contact center or telecom context, telephony integration handles session routing, dual-tone multi-frequency (DTMF) input, and media transport.

Note: DTMF refers to the tones generated when you press buttons on a telephone keypad. In the context of streaming and telephony context integration within voice agents, DTMF is used

Learn about

Understand memory-augmented agents and their capabilities (https://docs.aws.amazon.com/prescriptive-guidance/latest/agentic-ai-patterns/memory-augmented-agents.html)

Understand multi-agent collaboration patterns and use cases (https://docs.aws.amazon.com/prescriptive-guidance/latest/agentic-ai-patterns/multi-agent-collaboration.html)

Understand observer and monitoring agents for AWS systems (https://docs.aws.amazon.com/prescriptive-guidance/latest/agentic-ai-patterns/observer-and-monitoring-agents.html)

Understand components of LLM-augmented cognitive agents (https://docs.aws.amazon.com/prescriptive-guidance/latest/agentic-ai-patterns/overview-of-llm-augmented-cognition.html)

Understand the capabilities of Amazon Bedrock AgentCore (https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/what-is-bedrock-agentcore.html)

How to

as a signal input mechanism during a phone call, especially in interactive voice response (IVR) systems. DTMF inputs enable the agent to:

- Recognize menu selections (for example, "Press 1 for billing. Press 2 for support.")
- Collect numeric inputs (for example, account numbers, PINs, and confirmation numbers)
- Trigger workflows or state transitions in call flows
- Revert from speech to touch-tone when necessary

1. Reasons through LLM stream context
   - The query is sent to the agent, which passes it, along with any session metadata (for example, caller ID, prior context), to an LLM.
   - The LLM generates a response, possibly using a chain-of-thought strategy or multiturn memory if the interaction is ongoing.
2. Returns a voice response
   - The agent converts its response to speech using text-to-speech (TTS).
   - It returns audio to the user through a voice channel.

Capabilities

- Real-time speech understanding and generation
- Multilingual I/O with STT and TTS support
- Integration with telephony or streaming APIs
- Session awareness and memory handoff between turns

Common use cases

- Conversational IVR systems
- Virtual receptionists and appointment schedulers
- Voice-driven helpdesk agents
- Wearable voice assistants

Use AWS SDKs to perform batch and streaming transcriptions (https://docs.aws.amazon.com/transcribe/latest/dg/getting-started-sdk.html)

Use code examples to get started with Amazon Nova Sonic (https://docs.aws.amazon.com/nova/latest/userguide/speech-code-examples.html)

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- Voice interfaces for smart homes and accessibility tools

Implementation guidance

You can build this pattern using the following tools and AWS services:

- Amazon Lex V2 or Amazon Transcribe for STT
- Amazon Polly for TTS
- Amazon Chime SDK, Amazon Connect, or Amazon Interactive Video Service (Amazon IVS) for streaming and telephony
- Amazon Bedrock for reasoning with Anthropic, AI21, or other foundation models
- AWS Lambda to connect STT, LLM, TTS, and session context

(Optional) Additional enhancements may include the following:

- Amazon Kendra or OpenSearch for context-aware RAG
- Amazon DynamoDB for session memory
- Amazon CloudWatch Logs and AWS X-Ray for traceability

Summary

Speech and voice agents are intelligent systems that interact through natural conversations. By integrating speech interfaces with LLM reasoning and real-time streaming infrastructure, voice agents enable seamless, accessible, and scalable interactions.

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Nova › userguide

Using Amazon Nova as a foundation model in an AI agent (https://docs.aws.amazon.com/nova/latest/userguide/agents-use-nova.html)

Amazon Nova foundation model powers autonomous agents via Amazon Bedrock Agents, enabling API invocation handling, context memory management, knowledge base integration, and prompt

engineering customization. Invoke Converse API tool integration also possible.

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AWS discontinues support for Amazon Lex V1 on September 15, 2025. Document details slot properties like name, type, version, priority, utterances, and prompt.

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AWS Builder Center (https://builder.aws.com/)

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Helpful Links

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